

DATA 266 — Homework 2

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SID4 = 0670 | SEED = 670 | SLICE = 670 | HP_ID = 4 | CLS_A = 0 | CLS_B = 5

Git repo: <https://github.com/snehas-SJSU/Data266-0670/tree/main/HW-2>

HP_ID 4 is only reported. This homework has no capacity / lr / schedule arm, so I did not change chunk size, epochs, or model size from it.

Notebooks: embeddings.ipynb, rag.ipynb, optimizations.ipynb, amp_cuda_colab.ipynb.
Kernel: Python (HW1). Mixed precision ran on Colab Tesla T4.

1. Embedding transfer learning (4 points)

I loaded Google News Word2Vec (word2vec-google-news-300). 3,000,000 words, 300-d. Those vectors come from news, not movie reviews, so plot can look like a conspiracy and screen can look like a phone.

I copied the five word vectors first. Then I trained a Word2Vec model on 5,000 IMDB reviews (seed 670), after stuffing in the Google News vectors for words that overlap. That is the transfer step. `api.load` only gives KeyedVectors, which cannot `train()`, so I could not call `train()` on the downloaded object itself.

Words: cast, score, plot, screen, review.

1.1 Neighbors before and after

Table 1 — top-3 nearest neighbors (cosine)

Word	Before (Google News)	After (IMDB)
cast	casts=0.722; casting=0.719; Cast=0.664	casting=0.726; actors=0.706; supporting=0.700
score	scoring=0.720; scores=0.660; scored=0.638	scores=0.678; scoring=0.675; scored=0.618
plot	plots=0.762; Plot=0.652; plotting=0.633	plots=0.729; story=0.719; storyline=0.699
screen	screens=0.773; onscreen=0.612; LCD_screen=0.560	screens=0.691; stage=0.619; theatre=0.593
review	reviewed=0.663; reviewing=0.661;	reviews=0.784; uwe=0.694; seagal=0.687

reviews=0.638

plot and screen moved toward movie language. cast picked up actors / supporting. score barely moved — still the same word family (scoring / scores). review went to reviews plus IMDB junk (Uwe Boll / Seagal). That last one is ugly but it is what the 5,000 reviews did.

In this subset, plot showed up in 1015 reviews and score in only 178, which matches who moved.

1.2 Figure — t-SNE for plot

I picked plot because in news it often means a conspiracy (one neighbor was even Guy Fawkes) and in reviews it means the story.

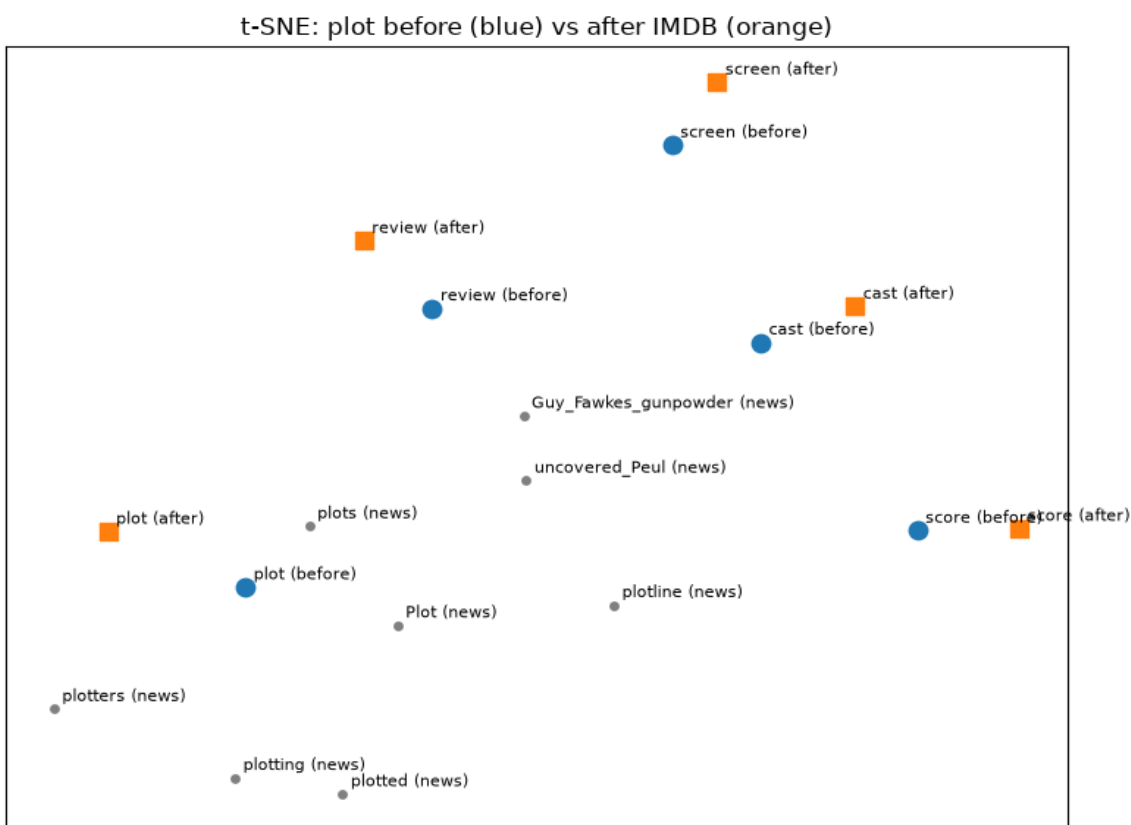


Figure 1. Blue circles = Google News. Orange squares = after IMDB. Gray = original news neighbors of plot. This is only a 2D picture. Distances here are not cosine in 300-d. Table 2 is the real shift number.

1.3 How far each word moved

Lower cosine = the vector moved more.

Table 2 — cosine(original, fine-tuned)

Word	cosine
review	0.7306
plot	0.7806
screen	0.7885
cast	0.8019
score	0.8618
• Shifted the most: review • Shifted the least: score	

2. RAG pipeline (4 points)

I loaded 10 Wikipedia movie pages with `WikipediaLoader`, split them (chunk 500, overlap 50), embedded with `MiniLM`, and stored them in an in-memory vector store. The pipeline is `PromptTemplate` + `retriever` + `Flan-T5-small`, called separately. I did not use one pre-built chain.

I picked colliding titles on purpose: *The Batman* / *Batman Begins* / *The Dark Knight*, and two *Dune* films. The other pages are *Inception*, *Interstellar*, *Parasite*, *Titanic*, *The Godfather*.

2.1 Five questions (chunk 500 / overlap 50)

Q	Question	LLM answer	Top-1 source
1	Who directed The Batman?	Matt Reeves	The Batman (film)
2	In what year was Dune released as a film?	2021	Dune: Part Two (talking about the 2021 film)
3	Which award did Parasite win at Cannes?	Academy Award for Best Picture	Parasite (2019 film)
4	Who composed the score for Inception?	Nolan	Inception
5	Who played Jack Dawson in Titanic (1997)?	Leonardo DiCaprio	Titanic (1997 film)

Full chunks are in `rag.ipynb`.

2.2 Chunk size change (Q1 and Q2)

I rebuilt the store with chunk **200** / **overlap 20** and re-ran Q1 and Q2.

- Q1 still answered Matt Reeves. Rank 1 stayed *The Batman*. Ranks 2 and 3 became **Batman Begins** (Nolan / Bale). Smaller chunks made the Batman name collision worse. The 500/50 run had all three hits on *The Batman*.
- Q2 still answered 2021. *Dune: Part Two* was still ranked high.

2.3 Retrieval scoring (top-3, 500/50)

I scored retrieval by hand. The LLM can be wrong even when the chunk is right.

Table 3

Q	Gold	In top-3?	Rank	Notes
1	Matt Reeves, The Batman (film)	Yes	1	right page
2	Dune 2021	Yes	1	year is in the chunks; rank 1 is the Part Two page
3	Palme d'Or at Cannes	Yes	1	Palme d'Or is in rank 1; LLM still said Oscar
4	Hans Zimmer	No	—	top-3 never name Zimmer
5	Leonardo DiCaprio as Jack Dawson	Yes	2	rank 1 is the Halifax grave labeled J. Dawson

Retrieval Success Rate = $4 / 5 = 0.80$

2.4 Two RAG failures

Failure A — Q4, relevant chunk not retrieved. Gold is Hans Zimmer. The retriever stayed on Inception but the top-3 chunks are the opening (Nolan, DiCaprio), a screenplay paragraph, and Oscars / box office. Chunk 3 says nominated for Best Original Score and never names Zimmer. I only kept the first 15,000 characters of each Wikipedia page, so the composer section may not even be in the doc. The LLM answered **Nolan**, which is sitting in chunk 1.

Failure B — Q3, context was right, LLM was not. Gold is Palme d'Or. Rank 1 has Palme d'Or and also Academy Award for Best Picture, with the Oscar sentence first. Flan-T5 answered **Academy Award for Best Picture**. That is not a Cannes award. The retriever did its job. The small LLM copied the Oscar.

3. Training optimizations (2 points)

Same 12-layer MLP, width 512, batch 64, 40 Adam steps, lr 0.001, seed 670. Device: **mps** (Apple GPU) for tensor-create, init, checkpointing, and accumulation. `mem_mb` there is MPS current allocated memory, not CUDA peak, so it barely moves. Mixed precision is a separate CUDA run on Colab Tesla T4 (`amp_cuda_colab.ipynb`).

Table 4

Tensor create, 2048×2048, mean of 3 after warmup, with `torch.mps.synchronize()`:

place	ms	mem_mb
cpu	61.087	n/a
mps	2.167	0.1

Weight init (same net / data / steps):

init	seconds	mem_mb	final_loss
default	4.559	28.4	1.4873
xavier	0.231	26.3	1.6886
zeros	0.204	26.3	2.2914

Checkpointing:

checkpoint	seconds	mem_mb	final_loss
False	0.273	26.3	1.4873
True	0.333	26.3	1.4873

Gradient accumulation (effective batch still 64, 40 opt steps):

accum	micro	seconds	mem_mb	final_loss
1	64	0.395	26.3	1.4873
4	16	0.990	26.2	1.5353

Mixed precision (Colab Tesla T4, `amp_cuda_colab.ipynb`; same net / data / batch / steps):

amp	seconds	peak_mem_mb	final_loss	forward dtype
False	0.2343	82.6	1.4924	torch.float32
True	0.2688	82.6	1.4793	torch.float16

Zeros init did not train (loss 2.29, about $\ln(10)$). Default vs Xavier both went down. I am not reading 4.56 s vs 0.23 s as “default is slow.” Default was the first train on a cold MPS graph.

Checkpointing took longer (0.273 → 0.333) and loss matched. Memory did not drop on this MPS meter.

Accumulation: loss stayed close (1.49 vs 1.54), time went up because each step does four forwards.

Mixed precision used real fp16 on CUDA (`torch.autocast(..., dtype=torch.float16) + GradScaler; probe used_fp16: True`). Warmup was untimed. Times are the mean of 3 with `torch.cuda.synchronize()`. AMP was a little slower ($0.2343 \rightarrow 0.2688$ s). CUDA peak memory stayed 82.6 MB. Loss stayed close (1.4924 vs 1.4793). This net is small; Tensor Cores did not help.

4. Short conclusion

Fine-tuning Google News on IMDB did move movie words. plot went toward story / storyline, screen toward stage / theatre, review moved the most (cosine 0.73). score moved the least (0.86) and still looks like sports morphology.

RAG retrieval was 4/5. The two misses I care about are Q4 (Zimmer never in the top-3, LLM said Nolan) and Q3 (Palme d'Or was in the chunk, LLM said Oscar anyway). Smaller chunks made Batman titles collide more.

On MPS, making a big tensor on GPU is faster than CPU. Checkpointing and accumulation behaved like the lecture on time. Memory did not drop on the MPS meter. Real mixed precision on a Tesla T4 used fp16; it did not speed this 12×512 MLP up, and CUDA peak memory stayed the same.